NeuroSync Diagnostics

Wearable Early-Detection Platform for Neurological Disorders

The Problem: Neurological Decline Is Diagnosed Too Late

Conditions such as early-stage epilepsy and mild cognitive impairment are typically diagnosed only after visible symptoms disrupt a patient's daily life, by which point treatment options are far more limited. Clinical EEG monitoring is expensive, requires a hospital visit, and captures only a short snapshot of brain activity — missing the subtle, intermittent patterns that precede diagnosis.

Our Solution: Continuous At-Home Neural Monitoring

NeuroSync has developed a lightweight, dry-electrode EEG headband paired with an AI analysis platform that continuously monitors brain activity during normal daily life. Our model flags early anomalous patterns and generates clinician-ready reports, enabling neurologists to catch warning signs months earlier than episodic in-clinic testing (B2B2C model through partner clinics).

Technology Advantage

- Custom dry-electrode hardware achieving clinical-grade signal quality without gel or a technician, validated against a 120-patient clinical pilot.
- On-device signal filtering paired with a cloud deep-learning model trained to distinguish artifact noise from clinically relevant anomalies in real time.
- HIPAA-compliant data pipeline with end-to-end encryption, built for direct integration into partner clinics' existing EHR systems.

Embedded C++ Firmware · PyTorch · HIPAA-Compliant Cloud Pipeline

$18B+ TARGET NEURO-DIAGNOSTICS MARKET	**3-6 mo** EARLIER DETECTION	**B2B2C** BUSINESS MODEL

Proposal for Horizon Health Partners

We are seeking a **$500,000 Seed** investment to complete FDA pre-submission requirements and expand our clinical pilot to five additional partner clinics.

This opportunity fits Horizon Health Partners' thesis of funding early-stage diagnostic technologies that shift neurological care from reactive treatment to proactive, data-driven early intervention.